3 **(a)** **(i)** Gravitational potential at a point is defined as **work done per unit mass** by an external agent to **move a test mass from infinity to that point** in the gravitational field (without a change in kinetic energy)

(ii) Gravitational potential at **infinity is defined as zero** and since gravitational force is attractive,

work done by external agent to move an object from infinity to a point in the gravitational field **is negative**, so potential is negative

(b) The **gradient of the potential–displacemen**t graph gives the gravitational **field strength, which is a vector**, thus, **the sign indicates the direction** of the gravitational field strength

Near the earth, the net gravitational field strength is directed towards the earth due to its larger influence than the moon while **near the moon, the net gravitational field strength is directed towards the moon** due to its larger influence. Hence, **their sign are opposite**

(c) **(i)** 1. Since the gravitational force the planet exert on moon A and B provides the centripetal force for them to rotate about the planet,

$$\frac{GM_pM_A}{r_A^2} = \frac{M_A v_A^2}{r_A} \text{ (together with statement for B1)}$$

$$\frac{GM_p}{r_A} = \frac{v_A^2}{1}$$

Since GM_P are constant, $v_A^2 \alpha 1/r$,

$v_A / v_B = (r_B / r_A)^{1/2}$

$= (2.2 \times 10^{10} / 1.3 \times 10^{8})^{1/2}$

$= 13$ (13.0)

Fractions not accepted

2. $v = 2\pi r / T$

$v \propto r / T$

$T_A / T_B = (r_A / r_B) \times (v_B / v_A)$

$= (1.3 \times 10^{8} / 2.2 \times 10^{10}) \times (1 / 13)$

$= 4.5 (4.54) \times 10^{-4}$

(ii) $T = 2\pi / 1.7 \times 10^{-4} = 3.70 \times 10^{4}$ s

$T_B = 3.70 \times 10^{4} / 4.54 \times 10^{-4}$

$= 8.1 \times 10^{7}$ s

4 (a) (i)